MTH105

Examination - January Semester 2025

Fundamentals of Mathematics

Monday, 14 April 2025 **04:00 pm - 06:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **SIX (6)** questions and comprises **FIVE (5)** printed pages (including cover page).

2. You must answer **ALL** questions.

3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.

4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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(Full marks: 100)

Question 1a

Employ a truth table to illustrate the statement

$$(p \to q) \wedge (\sim r \to q).$$

(7 marks)

Question 1b

Use a chain of logical equivalences to show that

$$(p \to q) \to (\sim p)$$

is logically equivalent to

$$\sim q \vee \sim p.$$

(6 marks)

Question 1c

Use the rules of inference to show that the following argument is valid.

$$p \to (q \to r) \text{ (premise)}$$

$$q \wedge \sim s \text{ (premise)}$$

$$r \to s \text{ (premise)}$$

$$\therefore \quad \sim p \text{ (conclusion)}$$

(7 marks)

Question 2a

Use mathematical induction to show that

$$\sum_{r=1}^{n} \frac{18}{(r+1)(r+3)(r+4)} = \frac{n(n^2+9n+17)}{(n+2)(n+3)(n+4)}$$

for all positive integers n.

(12 marks)

Question 2b

By using the fact that

$$\sum_{r=1}^{n} \frac{18}{(r+1)(r+3)} = \frac{3n(5n+13)}{2(n+2)(n+3)}$$

or otherwise, derive an expression for

$$\sum_{r=1}^{n} \frac{18}{(r+1)(r+4)}$$

in the form

$$\frac{n(An^2+Bn+C)}{2(n+2)(n+3)(n+4)},$$

where A, B, C are constants to be determined.

(6 marks)

Question 3a

Show that $\gcd(6528, 426) = 6$ and express 6 as an integral linear combination of 6528 and 426.

(10 marks)

Question 3b

Let $a = 6528 \times 17 \times 29$ and let $b = 426 \times 29$. State the value of $e = \gcd(a, b)$ and use the result found in Question 3a to express e as an integral linear combination of a and b.

(5 marks)

Question 4a

Let $\mathbb{N}^*$ denote the set containing the non-negative integers, and let $\mathbb{N}$ denote the set containing the positive integers.

Define a function $f : \mathbb{N}^* \to \mathbb{N}$,

$$f(x) = \begin{cases} 2x^3 + x + 1\ , & x \text{ odd} \\ x^2 + 2\ , & x \text{ even} \end{cases}.$$

Determine with justification where or not f is injective, and whether or not f is surjective.

(9 marks)

Question 4b

Define an equivalence relation

$$R = \{(a,a),(a,b),(b,a)(b,b),(c,c),(c,d),(d,c),\\ (c,e),(e,c),(d,d),(d,e),(e,d),(e,e),(f,f)\}$$

Describe the equivalence classes of R by listing down the elements of each of them.

(5 marks)

Question 5a

Define the universal set to be $S = \{x \in \mathbb{Z}\ :\ 1 < x < 26\}$. Define also the sets

$$A = \{y \in S : y \text{ is prime}\}, B = \{y\ \in S : y \text{ is a factor of } 24\},$$

$$C = \{y \in S : y = 4\text{k} + 1 \text{ for some integer k}\}.$$

Determine the elements of the following sets:

$$A \triangle B, \text{ and } (A \cup B) \cap C, \text{ and } S \setminus (B \cup C).$$

(9 marks)

Question 5b

Let A, B, C be non-empty sets. Show that

$$(A \triangle B) \setminus C = (A \setminus C) \triangle (B \setminus C).$$

(6 marks)

Question 6a

Provide the negation of the following statement:

$$\forall x \in \mathbb{Q}\ \exists y \in \mathbb{R}\ (x + y = 0) \vee (x = 0 \wedge y = 0).$$

(3 marks)

Question 6b

Give a counterexample to show that the following statement is false:

$$\forall x \in \mathbb{N}\ \forall y \in \mathbb{N}\ (x^2 + y^2 \text{ is prime}) \rightarrow (x + y \text{ is prime}).$$

(3 marks)

Question 6c

Suppose the ordered pair (a, b) has domain

$$D = \{(x, y)\ :\ x, y \in \mathbb{N},\ 1 \leq x \leq 6,\ 1 \leq y \leq 6\}.$$

Determine the truth set of the following predicate:

$$(\text{Either } a \text{ or } b \text{ is prime}) \rightarrow (10a + b \text{ is a multiple of } 7)$$

(5 marks)

Question 6d

Suppose that a, b, c, d are real numbers such that $ab + cd$ is rational. Use either the method of contraposition or the method of contradiction to show that if $ad + bc$ is irrational, then either $a - c$ is irrational, or $b - d$ is irrational.

(7 marks)

----- END OF PAPER -----